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RESEARCH

On-chip inductive sensor for ultra-high-throughput integrated detection of oil contamination and viscosity

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Abstract
A novel on-chip inductive detection sensor has been developed, offering a new method for analyzing contaminants and viscosity in hydraulic oil. An ultra-high-throughput microchannel with a rectangular cross-section has been designed, along with a dual-core coil resonant method to generate a large-scale magnetic field with high sensitivity on the chip. The inductive sensing unit consists of two symmetrically arranged rectangular magnetic core coils, creating a detection area with a high magnetic field strength. A rectangular microchannel with a cross-sectional area of up to 6 mm^2 passes between the two magnetic core coils. Compared to traditional micro-inductive sensors, the throughput increased by nearly 2 orders of magnitude, reaching 120 mL/h. Using the microchannel and resonance measurement method, we successfully detected 30 μm iron particles and 80 μm copper particles. Furthermore, we have established a model that correlates oil viscosity with its transit time through the microchannel. Through the inductance signal, we can determine the time it takes for the oil to pass through the coils and subsequently calculate its viscosity using our theoretical model. This method allows for the integration of inductive detection and viscosity measurement without the need for additional sensor. In the experiment, we measured hydraulic oils of different viscosities and compared the results with measurements obtained using a viscometer to verify the accuracy of the viscosity measurements.

Keywords Oil contamination · Inductive sensor · Viscosity measurement · Particle detection · Microfluidic chip

1 Introduction

Regarding machine as a whole part, oil (such as hydraulic oil and lubricating oil) function like the blood, playing a vital role as a key component. During the operation of machinery, hydraulic oil serves various functions, including

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power transmission, lubrication of moving parts to reduce friction, and cooling to prevent overheating. Most importantly, hydraulic oil carries wear debris, which can be used to characterize the wear rate and condition of the machinery. Monitoring the condition of the hydraulic oil provides valuable insights into the equipment's health, allowing for improvements based on the acquired data (Liu et al. 2015; Ren et al. 2018; Yu et al. 2018; Zhu et al. 2017). Therefore, hydraulic oil condition monitoring has become a common method for assessing machinery status and preventing failures (Feng et al. 2019). Currently, the laboratory testing, as the primary method for testing oil samples on mechanical equipment, is adopted to achieve testing oil samples comprehensively, including the analysis of physical and chemical characteristics. However, the process is time-consuming, requires multiple complex steps, and must be carried out by experienced technicians, Additionally, the high cost of the equipment further complicates the process (Fan et al. 2022; Murali et al. 2009). With the development of precision machinery, the demand for higher-quality and cleaner hydraulic oil has also increased (Zhang et al. 2012; R. T. Farrukh K et al. 2018). Therefore, rapid assessment of hydraulic oil conditions, encompassing factors such as particle contaminations and viscosity, is of great significance to the condition monitoring of mechanical equipment.

Traditional oil contamination detection methods mainly include acoustic detection, optical detection, capacitance detection, and inductance detection. The acoustic, optical, or capacitive detection techniques have high detection sensitivity, but the quality of the test sample can be easily influenced by external disturbances, and it is difficult to distinguish the material of fragments, leading to inaccurate measurements and incorrect results (Wang et al. 2020; Bordatchev et al. 2014; Xu et al. 2015; Chen et al. 2020). Inductive detection, recognized as a non-destructive testing method (Zhang et al. 2017; Zeng and Teng 2018; Du et al. 2015), stands out due to its advantages such as simple structure and strong resistance to interference. More importantly, it can differentiate between ferromagnetic and non-ferromagnetic metal particles based on changes in the magnetic field. Given these exceptional features, a significant amount of research has been conducted on inductive sensors, resulting in substantial improvements in recent years. However, the inductive detection method is limited to detecting only metal particles in the oil sample and cannot assess other parameters of the oil. Since viscosity is a key factor in determining the quality of hydraulic oil, detecting the viscosity of the oil sample would provide valuable insights into the oil's quality (Bie et al. 2019). Achieving simultaneous detection of both contamination and viscosity typically requires adding additional sensors, which increases the complexity of the detection system.

Currently, low sensitivity remains a significant drawback of induction detection, limiting their application. In order to improve the sensitivity and expand its utilization, various methods have been employed, such as structural optimization, circuit optimization, and the use of magnetic cores and resonant principles. Specifically, Ran et al. (Jia et al. 2019) used a series resonant structure combined with an amorphous core to alter the impedance of the excitation circuit, enabling detection of 120 μm ferromagnetic particles and 210 μm Non ferromagnetic particles. Liu et al. (Liu et al. 2022) designed a parallel resonant circuit optimized by adjusting capacitance for detecting wear debris in lubricating oil. This resonant circuit allowed for the detection of 25 μm iron particles and 66 μm copper particles. Li et al. (Li et al. 2023) developed a microwave rectangular cavity resonant sensor to monitor the state of lubricating oil. The sensor successfully detected non-metallic pollutants in lubricating oil and distinguished pollutants with different properties. Muthuvel et al. (Muthuvel et al. 2021) proposed an inductive detection sensor based on the principle of passive wireless LC, where the sensing coil was arranged inside the oil pipeline to monitor metal debris in the pipeline online. Feng et al. (Feng et al. 2021) designed a rotating static magnetic field induction sensor, consisting of cylindrical magnetic poles and a magnetic shell, which could detect 100 μm ferromagnetic particles by generating a high-gradient static magnetic field with rotational symmetry.

Further research showed that micro-inductive sensors, composed of inductive coils and microfluidic chips, dramatically improve detection sensitivity. Following that, by adding magnetic nanoparticles inside the solenoid coil, Bai et al. (Bai et al. 2020) used the micro sensor effectively detected 20 μm iron particles and 80 μm copper particles. Wang et al. (Wang et al. 2022) contributed to enhancing the sensitivity of micro-inductive sensors through studies on the mutual inductance of the coil and the high magnetic conductivity of permalloy. Additionally, Hu et al. (Hu et al. 2022) designed a high-throughput sensor with a dual excitation and multi-induction structure, which generated a strong symmetric gradient magnetic field between the excitation components to ensure higher sensitivity. Shi et al. (Shi et al. 2021) designed an impedance sensor with an integrated silicon steel strip to create a high-gradient magnetic field, allowing for the simultaneous detection of both inductance and resistance parameters. However, despite the improvements in sensitivity, the low throughput, usually with the diameter of the micro channel around 200–500 μm, is an obvious disadvantage of micro-inductive sensor.

For viscosity measurement, there are two main approaches. One method involves measuring the correlation between input and output (Hong 2010). Ultrasound, audio resonance, or rotational torque can serve as input indicators. The second method measures physical quantities such as

time, force, or torque after simulating the flow of lubricating oil to understand its characteristics (Bhattad 2023). Zhang et al. (Zhang and Yang 2022) established a mathematical model and designed a viscosity measurement device. A square microchannel structure built into the device was used to measure the movement time of the sample, and the viscosity of the sample was successfully obtained. In summary, current inductance detection methods have significant applications in multi parameter detection, however, the throughput of the highly sensitive micro-inductive sensors needs to be further improved, and here is currently no research linking inductance detection methods to viscosity detection.

This article proposes a micro-inductive sensor with high throughput, high sensitivity, and the capability to detect both contaminations and viscosity of oil samples. Specifically, permalloy sheets and LC resonant circuits are integrated into the square coil to enhance detection sensitivity, while the microchannel's cross-sectional area is expanded by 100 times to enable high-throughput detection. Additionally, a viscosity measurement model based on Stokes flow is developed within a rectangular microchannel, establishing the theoretical relationship between the kinematic viscosity of hydraulic oil and its flow time. As a result, the oil viscosity can be measured while simultaneously detecting the inductance signal, without the need for additional sensors or extra detection steps.

2 Materials and methods

2.1 Sensor design

As shown in Fig. 1(a), the microfluidic chip is designed to consist of two square solenoid coils and two permalloy sheets (magnetic cores). Each solenoid coil has two layers, with 20 coils per layer, tightly wound around the permalloy sheet. The diameter of the enameled wire is 100 μm, while the dimensions of the permalloy sheet are 2 mm in length, 30 mm in width, and 200 μm in thickness. The two magnetic core coils are symmetrically arranged, with a distance of 300 μm between them, and the edges of each core are aligned with the edges of the coils. The microchannel is designed in the area between the two coils, with a width (B) of 20 mm and a height (H) of 300 μm, as shown in Fig. 1(b) and Fig. 1(c). The two magnetic core coils are connected in parallel and oriented in the same direction, then connected to a capacitor in parallel as well. The detailed manufacturing process of the chip is provided in the Supporting Information.

2.2 Principle of particle detection

The theory of the inductive sensor used to distinguish between ferromagnetic and non-ferromagnetic particles is based on the principle of electromagnetic induction. In the alternating magnetic field generated by the coil,

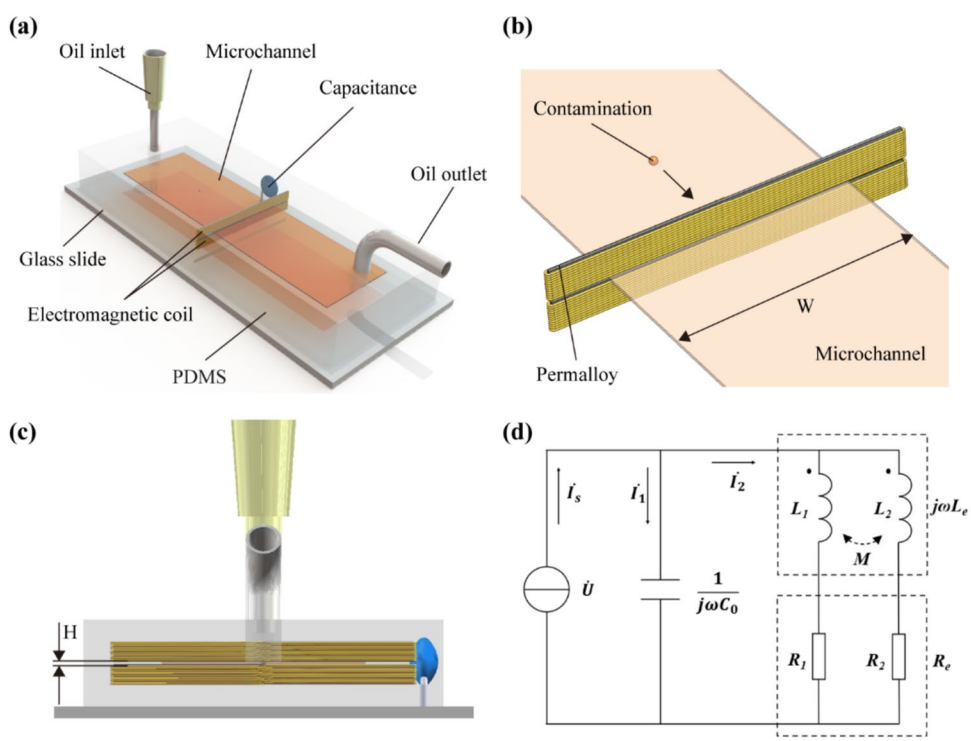

Fig. 1 **a** Overall design of the microfluidic chip; **b** enlarged view of the double-core coils; **c** side view of the microfluidic chip; **d** equivalent circuit of the double coil resonance

magnetization and eddy current effects occur within the metal particles, leading to a change in the coil's impedance through variations in the magnetic flux. The magnetization of ferromagnetic particles increases the magnetic flux, while the eddy current effects in non-ferromagnetic particles reduce the magnetic flux, resulting in signals with opposite directions.

In order to improve the detection sensitivity, a resonant circuit is adopted to enhance the magnetic field, as shown in Fig. 1(d), while the resonant frequency f_0 is

$$f_0 = \frac{1}{2\pi\sqrt{L_e C_0}}\sqrt{1 - \frac{C_0 R_e^2}{L_e}} \tag{1}$$

where C_0 is the capacitance value, L_e and R_e are the basic equivalent inductance and equivalent resistance, respectively. The equation shows that the signal generated by the particle is closely related to the frequency, and the maximum amplitude could be found near the resonance frequency (Huang et al. 2022).

2.3 Principle of viscosity measurement

According to the viscosity law, the flow rate of oil in a microchannel is correlated with its viscosity. When driven by a constant pressure pump, the time required for a fixed volume of oil to pass through the microchannel depends on its viscosity. The following assumptions are considered valid in this scenario: 1) The oil is incompressible; 2) Petroleum belongs to Newtonian fluid; 3) The microchannel is long enough and have a uniform width; 4) The flow rate remains stable; 5) The flow is laminar; 6) There is no slip at the gap walls (Zhang and Yang 2022).

Under constant-pressure pump pressurization, the oil flows uniformly through the microchannels, forming a sheet-like liquid column with a width of B and a height of H. Under steady-state conditions, when the pressure force and viscous force are balanced, the relationship between viscosity and time can be expressed by the following equation (detailed modeling process is shown in Supporting Information):

$$\upsilon = \frac{PH^2}{2\rho S^2}t \tag{2}$$

where υ is kinematic viscosity, P is the output pressure of the pump, H is the microchannel height, ρ is the density of the oil, S is the channel length, t is the flow time of the oil. The baseline of the inductance signal will change as the oil enters and exits the coil area, allowing the flow time of a given volume of oil through the microchannel to be directly determined from the inductance signal. Furthermore, based on the Walther's empirical equation (Bhattad 2023), the viscosity-temperature relationship for different oil samples can be calculated.

$$\lg(\lg(\upsilon + 0.7)) = 10.457 - 4.1008\lg(T) \tag{3}$$

In order to calibrate the oil viscosity measured at different temperatures, we calculated the viscosity variation curves for different hydraulic oils across various temperatures, as shown in Fig S2.

3 Results

3.1 Simulation results

A simulation model is established based on the chip design and theoretical analysis. Then the magnetic field generated by the double-core coils is calculated using COMSOL Multiphysics (COMSOL Inc.), the excitation voltage is 2 V and the frequency is 2 MHz. the results are shown in Fig. 2.

The simulation results show the magnetic flux density intensity along the X-axis and Z-axis. The X- axis represents the direction of the chip channel. Although the magnetic flux density in the middle is slightly lower than that of the permalloy part, it still reaches 10×10^{-5} T. The Y-axis represents the radial direction of the chip channel, where the magnetic flux density is also observed to be 10×10^{-5} T. This indicates that incorporating permalloy, a high-permeability material, effectively enhances the magnetic field intensity in the microchannel region. Additionally, to minimize edge effects, the microchannel should not be fabricated too close to the structure's edges. Therefore, the width of the microchannel is set to 20 mm, which is shorter than the 30 mm width of the permalloy sheet.

3.2 Experiment results

The detection system is shown in Fig. 3, which includes a constant pressure pump (FLUIGENT Flow EZ, FLUIGENT, France), an inductive sensor, a microscope (Nikon SMZ18, Nikon, Tokyo, Japan), a LCR meter (Keysight E4980A, Keysight Technologies Inc., Bayan Lepas, Malaysia), and a computer with LabVIEW software.

The hydraulic oil, containing both ferromagnetic and non-ferromagnetic particles, is driven by a constant pressure pump to flow through the inductive sensor. In order to excite the inductive sensor, the LCR meter with alternating current is used. Meanwhile, the LCR meter is used to collect real-time inductance data from the sensing unit and transmits it to a computer for analysis. For data processing, LabVIEW and MATLAB software are utilized. In the experiment, the voltage of the LCR meter is set to 2.0 V and the inductance

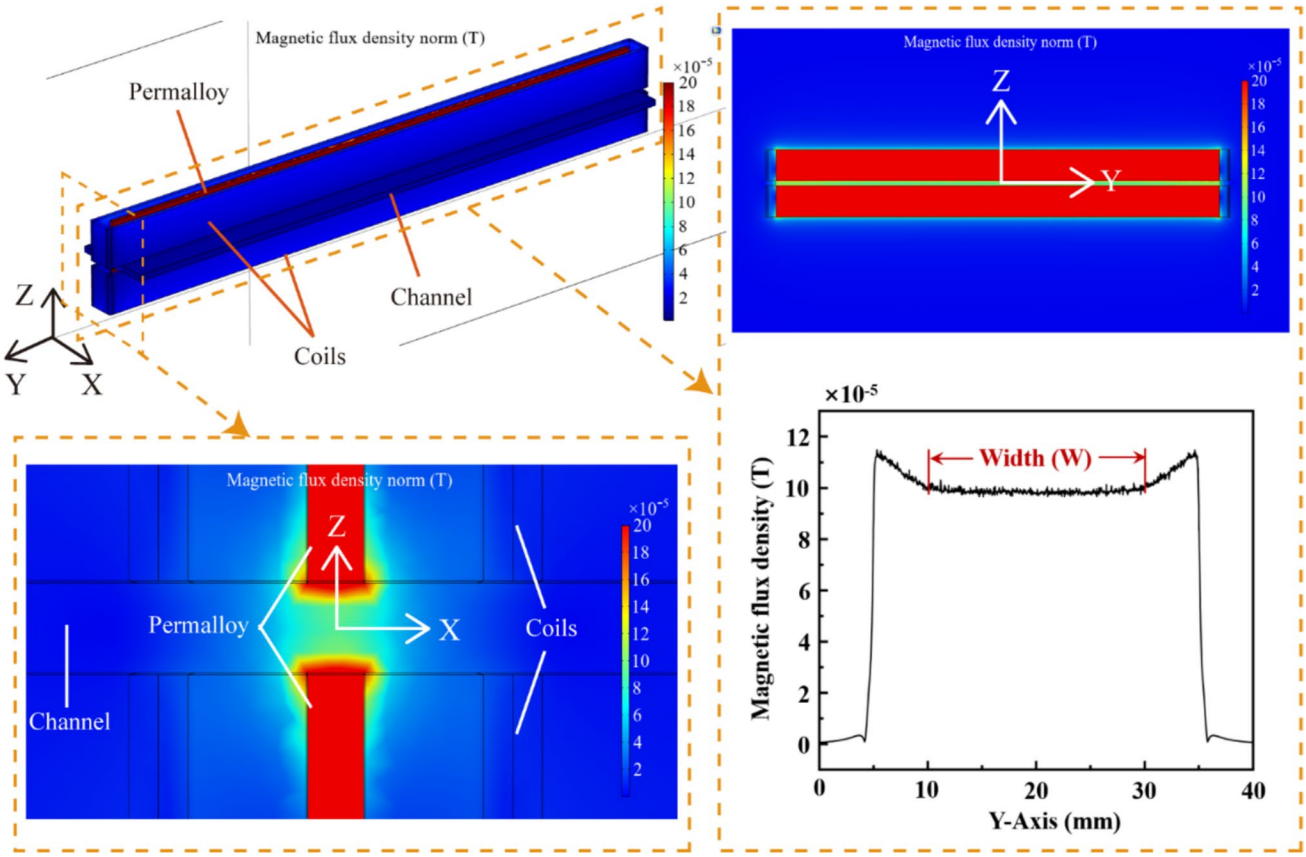

Fig. 2 Simulation results of the magnetic field generated by the double-core coils: the structural parameters of the simulation model are consistent with the chip design parameters (refer to Sect. 2.1), where the magnetic field curve represents the magnetic field intensity distribution along the Y-axis direction at the center of the two coil cores

sampling rate is set to 10 Hz, the pump flow rate is maintained at 2 mL/min.

3.2.1 Resonant frequency optimization

In order to prepare the oil samples, copper and iron particles are screened using different mesh sizes at first. Then, the iron and copper particles of varying sizes, each with a weight of 5 mg, are prepared using a precision balance (PRACTU224-1CN). Finally, two 5 mL portions of hydraulic oil (Great Wall L-HL46) are measured and poured into separate containers with iron and copper particles, respectively. An oscillator (IKA Vortex 3) is used to ensure the particles are evenly mixed before use.

To investigate the relationship between excitation frequency and inductance characteristics, 145 ~ 155 μm iron and copper particles are used for the experiment. The equivalent resistance and the basic inductance of the double-core coils are 228.51 Ω and 27.9 μH, respectively. According to Eq. (1), the resonant frequency of the detection circuit is calculated to be 1.605 MHz. This experiment detects the above two types of particles by changing the frequency in order to find an optimal excitation frequency near the resonant frequency. The results show that the signal noise at each excitation frequency remains essentially constant at 1.0×10^{-9} H. After comparing the variations in inductance amplitude and calculating the signal-to-noise ratio, the experimental results are shown in Fig. 4.

As shown in Fig. 4(a), when the excitation frequency is lower than the resonant frequency, iron particles increase the inductance, while copper particles decrease it. However, when the excitation frequency exceeds the resonant frequency, the opposite effect occurs: iron particles decrease the inductance, and copper particles increase it. At a frequency of 1.5 MHz, the inductance change caused by iron particles reaches its maximum, whereas the peak for copper particles occurs at 1.7 MHz. Based on the experimental results, it is concluded that when iron particles and copper particles pass through the detection area, the resonant frequency of the circuit would be impacted. Specifically, the resonant frequency decreases slightly when iron particles pass through the coils, while it increases when copper particles pass through. As shown in Fig. 4(b), when the resonant frequency is around 1.6 MHz, the signal-to-noise ratio for

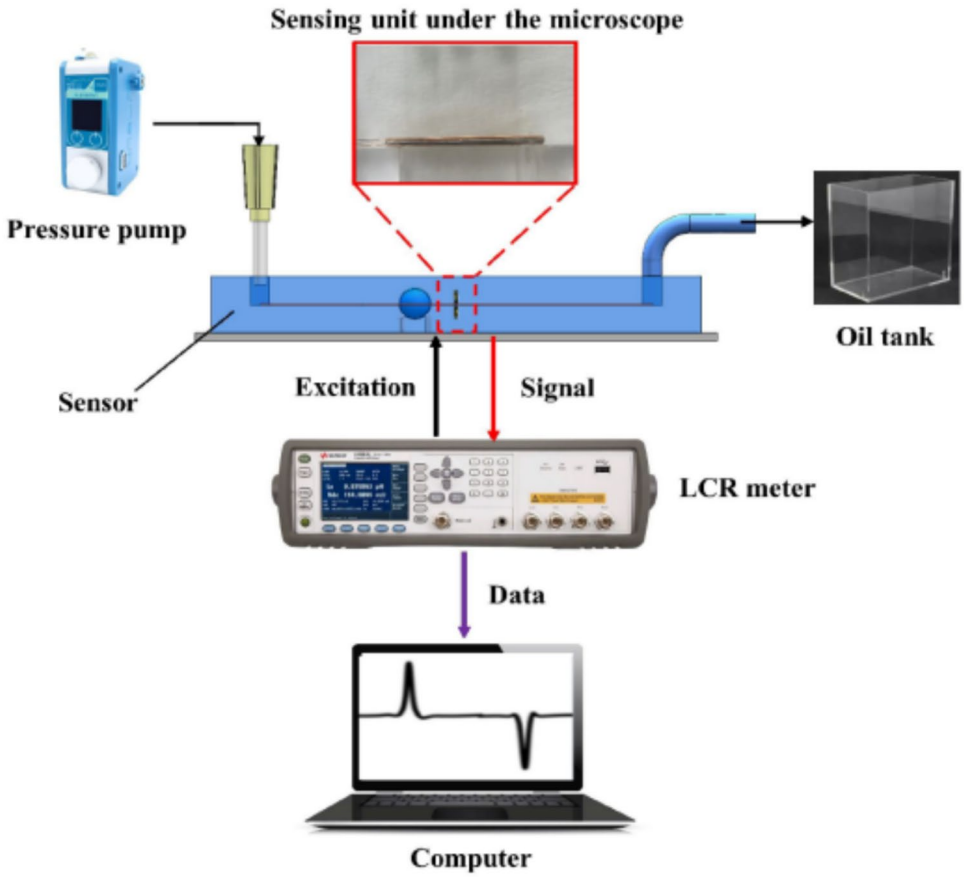

Fig. 3 Diagram of detection system

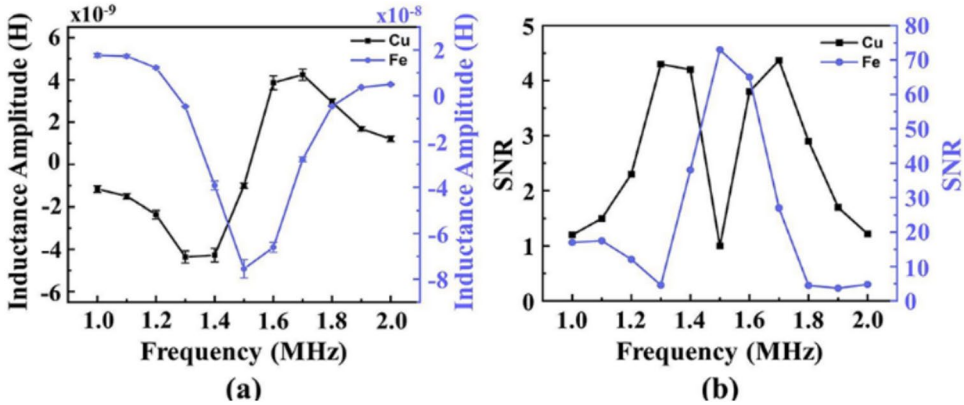

Fig. 4 **a** Detection results at different frequencies; **b** signal to noise ratio at different frequencies

iron particles increases by up to 73 times, while there is no significant change in the signal-to-noise ratio for copper particles. To effectively distinguish iron particles from copper particles while maintaining a high signal-to-noise ratio, 1.6 MHz is selected as the excitation frequency for resonance in the next experiments.

3.2.2 Comparative experiment between resonance and non-resonance

The previous research results (Zhang et al. 2022) has shown that the detection effect of a simple non-resonant coil improves as the frequency increases within a certain range. Therefore, a frequency of 2.0 MHz was chosen to excite the simple non-resonant magnetic core coils. In order to compare the detection effects with and without resonance, 145–155 μm iron particles and copper particles are used in

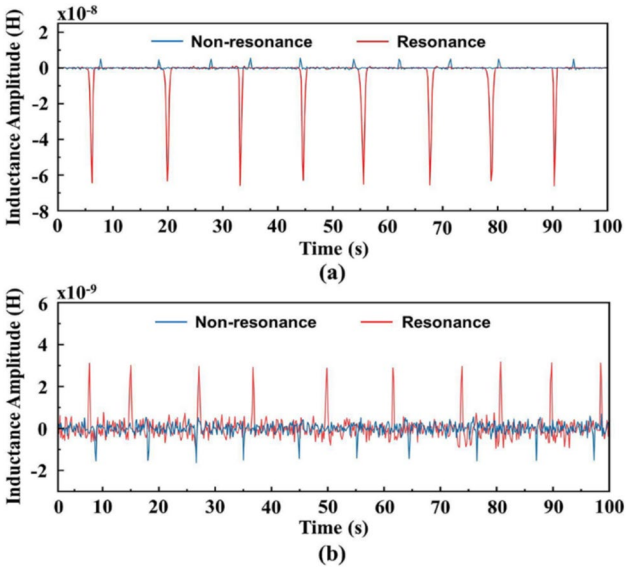

Fig. 5 Comparison of resonant and non-resonant detection results: **a** 145 ~ 155 μm iron particles; **b** 145 ~ 155 μm copper particles

the experiments. The results of the comparative experiment are shown in Fig. 5.

The comparison of resonant and non-resonant detection results is shown in Fig. 5(a) and Fig. 5(b). The noise level of resonant detection is similar to non-resonant detection. However, for metal particles of the same size, the amplitude of the resonant detection is significantly higher than the non-resonant detection, especially for iron particles, where the amplitude of resonant detection is 10 times higher than that of non-resonant detection. Based on these results, it can be concluded that the double-core coils resonant detection chip provides a better detection effect.

3.2.3 Relationship between particle size and detection amplitude

In order to calibrate the signal amplitude generated by particles of different sizes, iron and copper particles of varying sizes were detected under the optimal excitation frequency separately, as shown in Fig. 6(a) and Fig. 6(b). The results clearly show a positive correlation between the inductance detection amplitude and particle size. Finally, we successfully detected 30 μm iron particles and 80 μm copper particles, as shown in Fig. 6(c) and Fig. 6(d), the experiments were conducted with only iron particles and only copper particles, respectively.

3.2.4 Contamination differentiation detection

The above detection results indicate that the chip designed here exhibits high sensitivity. In order to further evaluate its performance, the sensitivity of the chip would be validated under high-throughput conditions. In the next step, oil samples containing iron and copper particles of varying sizes were injected into the chip using a pressure pump at a flow rate of 2 mL/min for contamination differentiation detection. The detection results are shown in Fig. 7.

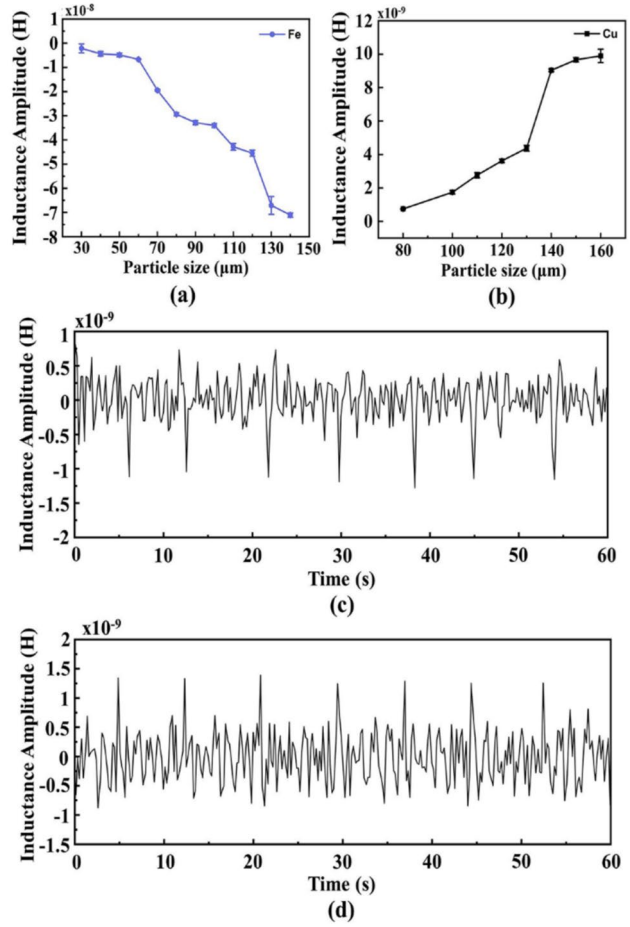

Fig. 6 Detection results at different particle sizes **a** Detection results of iron particles; **b** Detection results of copper particles **c** Detection limits of iron particle (30 μm iron particle only in the experiment); **d** Detection limits of copper particle (80 μm copper particle only in the experiment)

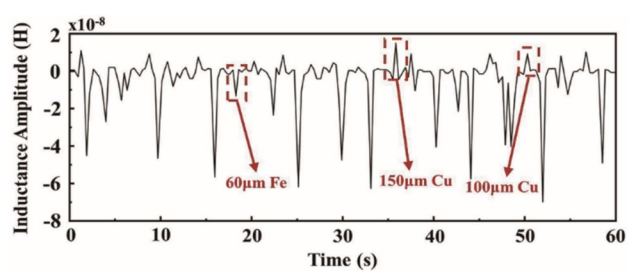

Fig. 7 Detection results of mixed contaminations

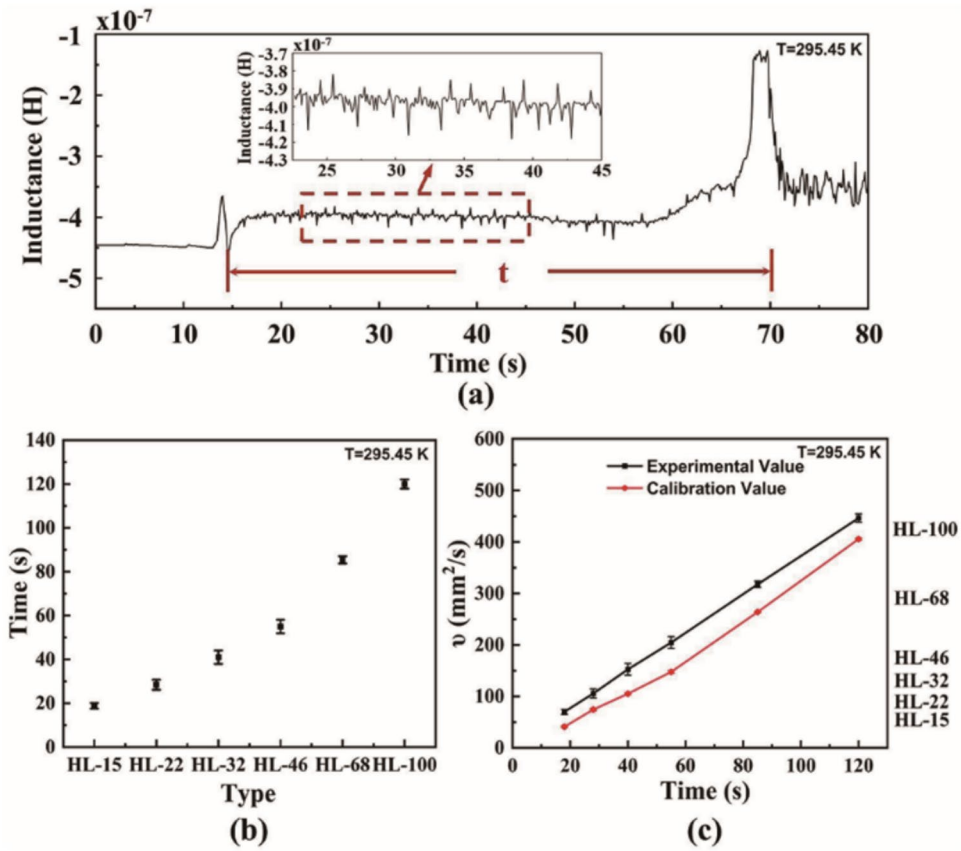

Fig. 8 **a** The signal of 2 mL oil sample with particles flowing through the chip; **b** The flow time of different oil samples; **c** Comparison of experimental values and calculated values

The detection results show that the chip can distinguish between iron and copper particles of varying sizes based on the direction of the signal, while also maintaining the high sensitivity mentioned earlier.

3.2.5 Viscosity measurement

In the viscosity experiment, we first take a 2 mL oil sample and used a pressure pump to completely inject the oil into the chip at a constant pressure of 80 mbar and a temperature of 295.45 K. By analyzing the change in the baseline of the inductance signal, we were able to determine the time it took for 2 mL of different types of oil to pass through the microchannel under constant pressure, as shown in Figs. 8(a) and (b). Using the viscosity measurement calculation model (Eq. (2)), we then obtained the relationship between the viscosity and flow time of different types of hydraulic oil at 295.45 K, as shown by the experimental values in Fig. 8(c). Meanwhile, we also measured the viscosities of the different oils using a Pinnacle viscometer (1833, Taizhou Jiaojiang Huanguang Glass Instrument Co., Ltd.) and compared them (calibration values in Fig. 8(c)) with the experimental results from the chip. The principle of the Pinnacle viscometer is based on Poiseuille's law. The viscosity of the liquid is calculated by measuring the time it takes for the liquid to flow through a capillary under the action of gravity.

To further compare the data, we used Pearson correlation coefficients to assess the degree of correlation between the viscosities obtained through the two different methods. Notably, there is a high correlation coefficient of 0.99045 between the experimental values and the calibration values (Fig. S3). This strong correlation further demonstrates the close relationship between the viscosities measured by the two methods.

The viscosity measurement results show a slight deviation between the experimental value obtained from the chip and the calibration value from the viscometer, with the chip's experimental value being slightly higher. This deviation is due to the length of the pipeline between the constant pressure pump and the chip inlet introduces flow resistance, resulting in a pressure drop in the fluid. This pressure drop increases the flow time of the oil, causing the viscosity calculated using Eq. (2) to be slightly higher than the value measured by the viscometer. For this reason, we conducted experiments to evaluate the impact of pipeline length on the flow time. The results, shown in Fig. S4, indicate that as the pipeline length increases, the flow time of the oil also increases accordingly. However, there is a strong linear relationship between the time it takes for the oil to pass through

the chip and its viscosity. Therefore, we can align the chip's calculated value with the actual measured value by applying a simple linear calibration (adding a correction coefficient).

4 Conclusions

This paper proposes a high-throughput, high-sensitivity multi-parameter oil detection method, enabling the integrated detection of oil particle contaminants and viscosity using a single micro-inductive sensor. First, a high-throughput microchannel with a rectangular cross-section is designed. By integrating the double-core coils and a resonant circuit, the system ensures high detection sensitivity while significantly enhancing oil detection throughput. Compared with traditional microchannels, its throughput is increased by 100 times, reaching 120 mL/h, and it has successfully detected 80 μm copper particles and 30 μm iron particles in oil. Meanwhile, a viscosity measurement model is developed using the rectangular cross-section microchannel, based on the relationship between oil pressure and flow time within the microchannel. The flow time, obtained directly from the inductance detection signal, is used to determine oil viscosity. Viscosity measurement experiments were conducted on different types of oil, and the results were compared with viscometer measurements to validate the model's feasibility. It is worth noting that although the inductive sensor significantly improves the oil detection throughput, it also results in an extended sensing area. As a consequence, to avoid multiple particles passing through the sensing region simultaneously and generating overlapping or interfering signals, the particle detection function of this sensor is not suitable for scenarios with very high particle concentrations. In such cases, the oil may need to be diluted prior to measurement to ensure accurate detection. In summary, the proposed multi-parameter oil detection method extracts critical information—including particle count, particle size, and oil viscosity—from a single inductance detection signal. This approach holds great promise for future applications in oil condition monitoring for machinery and other equipment.

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Author contributions Hongwei Guan: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal); Jian Feng: Data curation (equal); Formal analysis (equal);Investigation (equal); Methodology (equal); Writing – original draft (equal); Qingyi Cai: Data curation (equal); Investigation (equal); Yi Yang: Investigation (equal); Validation (equal); Chao Liu: Investigation (equal); Data curation (equal); Duo Sun: Formal analysis (equal); Validation (equal); Jienan Shen: Conceptualization (equal); Methodology (equal); Supervision (equal);Visualization (equal); Hongpeng Zhang: Conceptualization (equal); Methodology (equal); Supervision (equal); Funding acquisition (equal); Lin Zeng: Conceptualization (equal); Methodology (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal); Hui Yang: Conceptualization (equal); Methodology (equal); Funding acquisition (equal); Project administration (lead); Supervision (equal); Writing – review & editing (equal).

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Data availability The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Declarations

Conflicts of interest The authors declare no conflict of interest.

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